Solution Requirements

Date	3 October 2026
Team ID	SWTID-2026-5338
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/ Low)
1	Authentication	Users register and log in with an email and password (Google sign-in optional). Passwords are stored only as salted hashes, sessions use expiring tokens (JWT), and a password-reset link is sent by email. Unauthenticated visitors cannot generate or save comics.	High
2	Authorization levels	Guest: can view the landing page and sample comics only. Registered User: can create stories, generate, edit, save, export and share their own comics, and can access only their own data. Admin: can manage user accounts, monitor usage and API quota, review reported content and configure system settings.	High
3	External interfaces	Google Gemini API: text model to generate the story script, scene breakdown and dialogue; image model to generate panel artwork. Cloud database (user accounts, stories, comic metadata) and file/object storage for panel images. Email service for verification and password reset. PDF/image export library for downloads.	High

S.No	Requirement Category	Requirement Description	Priority (High/Medium/ Low)
4	Transactions processing	Each generation request is validated, converted into a structured prompt, sent to the Gemini API, and the response is checked, stored against the user's account and returned to the UI. Failed or timed-out calls are retried up to 3 times with a clear error message. Edits and saves are written atomically, and every generation is counted against the user's quota.	High
5	Reporting	Users: a "My Comics" dashboard listing saved comics with title, creation date and status, and export of any comic as PDF or PNG/JPG. Admin: usage report showing users, comics generated, Gemini API calls, error rate and quota consumption.	Medium
6	Business rules, etc.	A story prompt (idea, genre, art style, number of panels) is mandatory; panel count is limited to 4 to 12 per comic. Character descriptions are reused in every panel prompt to keep characters visually consistent. Unsafe or inappropriate prompts are blocked by content filters. Each user has a daily generation limit. Any single panel can be regenerated without redoing the whole comic. Users own the comics they create.	High
7	Compliance to laws or regulations	Follow data privacy principles of GDPR and India's DPDP Act 2023: user consent, collection of minimum data, and the right to access and delete account data. Comply with Google Gemini API terms and usage policies. Label content as AI-generated and do not imitate copyrighted characters or specific living artists.	Medium
8	Other	Edit panel text and speech bubbles, reorder or delete panels, save drafts and resume later, share a comic through a link, and use a responsive web interface on desktop and mobile.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Fast page loads and quick AI responses, with a visible progress indicator while content is being generated.	Page loads in < 3 seconds; story script generated in ≤ 10 seconds; each panel image in ≤ 20 seconds

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
2	Scalability	Stateless backend that can scale horizontally, with a request queue to manage Gemini API rate limits as users and stored comics grow.	Supports up to 1,000 concurrent users and 100,000 stored comics
3	Security & Data Privacy	All traffic over HTTPS; passwords hashed; data encrypted at rest; Gemini API key kept on the server and never exposed to the browser; input validation and rate limiting to prevent abuse; regular vulnerability checks.	TLS 1.2+ on all connections; AES-256 encryption at rest; no API keys in client code; no critical OWASP Top 10 issues
4	Reliability & Availability	Graceful handling of Gemini API errors with automatic retries, auto-save of work in progress, and regular backups of user data.	99.5% uptime per month; up to 3 automatic retries; daily backups
5	Usability & Accessibility	Simple guided flow (idea, generate, edit, export), clear labels, keyboard navigation, readable contrast, and a layout that works on phones, tablets and desktops.	Complies with WCAG 2.1 AA standards; a new user creates a first comic in under 5 minutes without help
6	Other	Maintainability and portability: modular, documented code with separate frontend, backend and AI-service layers, and browser-independent behaviour.	Works on the latest two versions of Chrome, Edge, Firefox and Safari; setup documented in README